



Article

Identifying rhetorical devices in Spanish and Catalan literature

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1. Introduction

This study used AI and machine learning to identify rhetorical language in ancient Spanish and Catalan literature. The text was written by three 15th-century female Iberian writers from Castilla Leon and Aragon. The Iberians were considered to be all the people living on the eastern and southern coasts of the Iberian Peninsula in the 6th century BCE. They were known for their distinct culture, craftsmanship, and agricultural techniques. In this project, we were provided with three XML files containing ancient scriptures. Each of the files has been partially labeled and annotated with tags. Our group focused on using machine learning to identify the following rhetorical devices for the rest of the text: metaphor, anaphora, pathos, and parallelism.

2. Research Statement

In this research, we aim to utilize natural language processing and large language models to identify rhetorical devices in three ancient Spanish and Catalan literary works. We strive to investigate whether machine learning and AI are suitable applications for this specific research. This article tracks our learning and research process from preprocessing data to training models and eventually with accuracies and results.

3. Common Preprocessing

3.1. Parsing XML files for Rhetorical Devices

We created a function that iterates through each <seg> element, checks for the type and subtype attributes, and extracts the text between them. For example, when parsing the files for metaphor, if the text is tagged as a metaphor (i.e., subtype="metaphor"), it adds that text to a list of metaphorical devices. Otherwise, it adds it to a list of non-metaphorical devices.

3.2. Tokenization

We loaded a Spanish SpaCy model "es_core_news_sm", which would break the text into smaller chunks(tokens).

3.3. Part of Speeches (POS) feature extraction

These features are essential for training the model, especially for rhetorical devices. We used the SpaCy library to import a function that analyzes every word and finds the appropriate part of speech for that word. While identifying

rhetoical language, it is important to have an adequate amount of information about the text. Finding the POS helps provide that extra information.

4. Rhetorical Devices & Models

4.1. Metaphor

Metaphor is a figure of speech where a word or phrase is applied to an object or action to which it is not applicable. I used OpenAI to generate 100 additional metaphor examples in Catalan to balance the dataset for metaphor and non-metaphor examples.

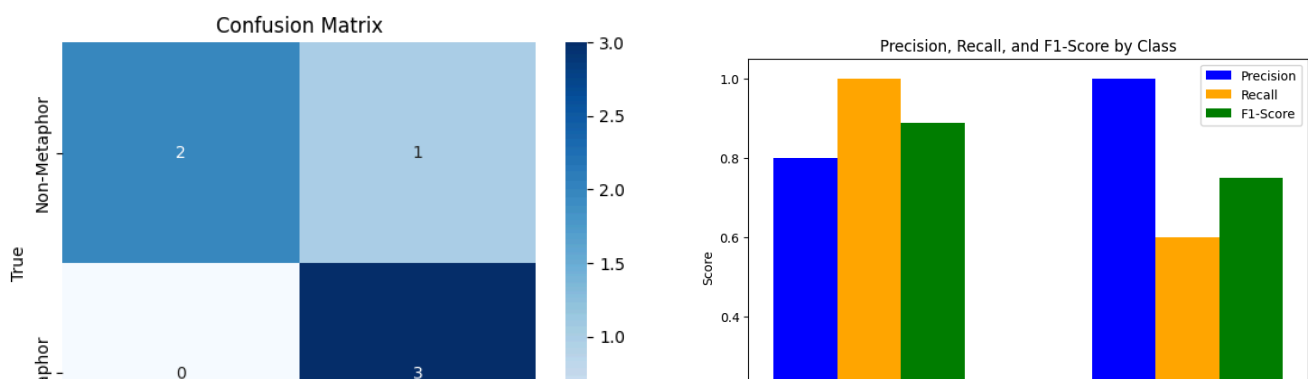
After preprocessing the metaphor and non-metaphor text (tokenization, lemmatization, and POS tagging) and assigning labels, the data was split into three parts: 80% for training, 10% for validation, and 10% for testing.

For metaphor classification, I used a pre-trained Spanish BERT model ([dccuchile/bert-base-spanish-wwm-cased](#)). BERT leverages self-attention to analyze relationships between words in a sentence. During pre-training, BERT learns general Spanish language patterns through two tasks:

1. Random words in a sentence are replaced with a [MASK] token, and the model predicts the original word.
2. The model predicts which sentence logically follows another.

Once pre-trained, I fine-tuned the model using the training set for binary metaphor classification, where metaphors are labeled as 1 and non-metaphors as 0. Through iterative training, the model optimized its weights using the loss function to minimize errors between predicted and actual labels.

Finally, the model's performance was evaluated on both the validation and test sets. The model achieved an accuracy of 84.6% on the test set, with strong precision for metaphors (100%) but lower recall (60%), meaning the model is very cautious in predicting metaphors, as everything it identifies is correct.



Finally, for the prediction section, I input all three texts from the works of Iberian women writers ("Memorias," "Vita," and "Arboleda"). The data was cleaned by removing numbers, unwanted symbols, and extra spaces. After cleaning, the text was split into individual sentences, which were then tokenized using the pre-trained Spanish BERT model for metaphor classification. Each sentence was classified as either a metaphor or a non-metaphor, and the sentences identified as metaphors were saved along with the source file information in a CSV file.

```
● Admin@Carmens-MacBook-Air AI-ML-Camp % /usr/local/bin/python3 /Users/Admin/AI-ML-Camp/AI-ML-Camp/scripts/Identifying_new.py
Number of metaphors detected: 156
```

For example, the first sentence we identified is "La libertad de decir y la originalidad de esta autora son fruto de la escuela del dolor," translated - "The freedom to speak and the originality of this author are the fruit of the school of pain." "Fruit" is metaphorically used to represent the result or outcome of something.

4.2. Anaphora

Anaphora, specifically rhetorical anaphora, is the repetition of a word or phrase at the beginning of neighboring clauses.

There were no available datasets online that consisted of rhetorical anaphora examples in Spanish. Additionally, there were only 9 total labeled instances of anaphora in the original dataset provided. So, I used OpenAI to generate 200 examples - 100 anaphora and 100 non-anaphora. Those 200 examples were my training data, while the Arboleda was used as testing data.

First, I parsed through the XML file to sort into anaphora and non-anaphora lists by finding when `seg_type` is "rhetoric" and `seg_subtype` is "anaphora." Everything that did not match the requirements was labeled as non-anaphora. The text and labels are saved in new variables for later use.

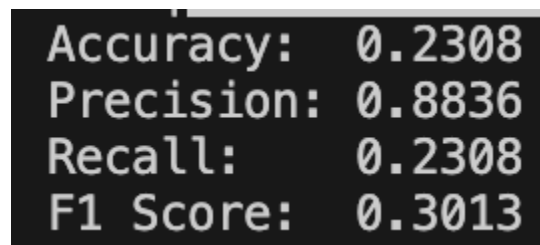
I used the HuggingFace datasets and the BETO model in my code. BETO is the Spanish version of BERT (a Natural Language Processing machine learning model) and is uploaded to the code using

`"dccuchile/bert-base-spanish-wwm-cased"`

It has a pretrained model, which I use to tokenize both my training and testing data. One issue I ran into was that BERT had to have `use_safetensors=True` for security reasons. Once that is complete, I convert my data to Pytorch tensors so my results are easier to print out.

Then, I start training my data using Trainer for five epochs. Lastly, the model is saved for later use.

Finally, in the prediction section, my model was tested on Arboleda, and the accuracy results are below. Even though the result isn't the best, I can continue building on it.



```
Accuracy: 0.2308
Precision: 0.8836
Recall: 0.2308
F1 Score: 0.3013
```

4.3. Pathos

Pathos is when the author purposefully evokes a strong emotion in the readers by using some emotional words or descriptions.

- **Failed methods:** I tried to use NLP models like BERT models to identify other examples of pathos existing in the documents. However, after carefully analyzing the three XML files, I found out that there are only 13 examples. Due to this limited number of examples, there is not enough preprocessed data to train and test the model. Additionally, since the text is written in Castilian Spanish and Catalan, which are very different from regular Spanish, there are no easily accessible datasets on the internet for us to use. Therefore, this method has to be passed.
- **LLMs:** LLMs allow users to do zero-shot learning or few-shot learning, which is perfect for minuscule datasets, like in my case. Therefore, I decided to implement prompt engineering through LLMs, and the model we chose is Gemini 1.5 flash. To implement this, I told Gemini its tasks and role, and input 4 pathos examples that Holy prelabeled from the XML files. I bring in Gemini using a generated API key and feed a part of the text that is unlabeled to the model. To check the accuracy of the model, I mixed 5 other examples that are prelabeled in the XML files with other

irrelevant text from the documents and fed them into the model to see if the model predicted them as pathos. The model correctly identifies all five pathos, even though it also identifies one more that is not supposed to be pathos. Thus, I think the model is pretty accurate. Below is the prompt that I gave to Gemini:

```
You are an expert in rhetorical analysis.

The following text is written in Spanish or Catalan and may contain multiple examples of emotional appeals (pathos).
Your task is to:

1. Read the full text.
2. Identify all specific segments that use emotional language to appeal to the reader's feelings.
3. Extract and list each of those emotional segments (in their original language).
4. For each, explain briefly in English why it is an emotional appeal.

Here are a few examples of emotional text:

1. "el qual conpuso Teresa de Cartajena seyendo apasyonada de graues dolencias"
   This is an example of pathos.

2. "Salid, señores, yveréis la más desventurada, desamparada y más maldita mujer del mundo"
   This is an example of pathos.

3. "Y yo le dije, con muchas lágrimas"
   This is an example of pathos.

4. "Los que morauan en tinieblas y en sonbra de muerte, luz les es demostrada."
   This is an example of pathos
```

Because of the quota limitation from Gemini, we were not able to input the whole text. However, we fed it some chunks of the document, and it works properly.

4.4. Parallelism

Parallelism is a rhetorical device that involves having a similar grammatical structure. A famous example of parallelism is the line “I came, I saw, I conquered.” The grammatical structure for that is as follows: pronoun+noun, pronoun+noun, pronoun+noun. The repetition of the pronoun and noun is the deciding factor on whether it is considered parallelism or not. There were no examples of parallelism in the files that were provided, so there is no solid accuracy number. However, for my code, I started by using the “spacy” library in order to retrieve the parts of speech for each word. I later realized that the library was having trouble processing the ancient language. To fix this, I found a dictionary that would translate the ancient words into their modern-day terms. In order to figure out if the line used parallelism or not, I used N-grams. N-grams are used to identify how many times a word, or set of words, appears within a text. I used N-grams to find how often a set of the parts of speech occurred. If the set was recurring throughout the line, I would print out the line as parallelism. I later tested several of the sentences through ChatGPT, and it declared many of those examples as accurate. Even though that method is not optimal, it helped give an understanding of how well the model worked.

5. Discussions and Limitations

The biggest challenge and limitation that we have is the limited amount of data and the rarity of old Spanish being used or trained by machine learning models. With the limited amount of data we have, we could not use many of the readily available machine learning

models. Therefore, the models also struggled to translate the ancient Spanish. Developing more adaptable models for ancient literature might be an interesting topic to focus on for future studies.

As with any complex field, NLP comes with its fair share of challenges. The computational complexity of NLP tasks can be a significant limitation, for instance. Processing large amounts of text data requires substantial computer power and time, making it challenging to achieve real-time or near-real-time analysis. Moreover, improving the efficiency and speed of NLP algorithms is another ongoing challenge. However, challenges present exciting opportunities for innovation and growth. Additionally, there was also a chance that models would immediately assume that there was a connection with certain words, without checking if it was true or not. Furthermore, our project only checked for four rhetorical devices. In reality there were probably many more rhetorical devices within the text, but we did not have the time or resources to find them.

On the other hand, large language models also have their limitations. For example, it is really hard to track and assess the accuracy of these models, as there is no clear way to get an accuracy score. Therefore, all we did is to access the output from the model by reading it. This is not the most ideal method and could lead to potential errors in the output. Nonetheless, LLMs do seem like the most optimal method to use for more complex rhetorical devices, considering the uniqueness in the language and the limited data in this project.

6. Conclusion

In conclusion, our team believes that machine learning could help improve the efficiency of this research to some extent, but the accuracy of the outputs should not be trusted completely. We suggest using large language models for this specific task due to the very limited amount of data provided. Furthermore, the four rhetorical devices that we focused on in this research are only the ones that have the most data available or can be easily identified. Therefore, some of the more complicated devices might need to be done manually. Lastly, we look forward to seeing more mature models for processing ancient literature in different languages emerge in the future.